

TILLAGE AND LAND DEVELOPMENT EQUIPMENT

2.1 Tillage — Definitions

Tillage is the mechanical manipulation of soil to provide a favourable environment for crop growth.

- **Primary tillage:** initial, deeper opening of soil (12–30 cm or more) — MB plough, disc plough, chisel plough, subsoiler, rotary plough.
- **Secondary tillage:** shallower, refining operations (5–10 cm) — harrows, cultivators, levellers, pulverizers, rollers.
- **Minimum tillage:** least soil manipulation necessary; **Zero/No-till:** direct seeding into unprepared soil; **Conservation tillage:** ≥30% of surface covered by residue after planting; **Strip/Rotary/Mulch tillage** are other classes.

2.2 Mouldboard (MB) Plough

Performs cutting, lifting, **inverting (complete inversion is its distinguishing action)** and pulverizing the furrow slice. Good for burying weeds and trash.

Parts: share (penetrates and cuts), mouldboard (lifts and inverts), landside (absorbs side thrust; rear end has heel), frog (base to which all parts bolt), shin (cutting edge extension of MB), tail piece.

- **Share types:** slip share, slip-nose share, shin share, **bar-point share** (most common where abrasive soils rapidly wear points; point can be extended).
- **Share materials:** chilled cast iron or soft-centre steel (outer hard layers over soft core).
- **Mouldboard types:** *General purpose* (medium pitch, most common), *Stubble* (short, abruptly curved — maximum pulverization), *Sod/Breaker* (long, gently curved — complete inversion without breaking slice), *Slat* (sticky soils), *High-speed* (slender, for 6–8 km/h+).
- **Clearance/suction:** *Vertical suction (down suction)* helps penetration (typically 3–5 mm for tractor ploughs); *Horizontal suction (side suction/land suction)* helps maintain furrow width (typically 3–10 mm).
- **Plough size** = width of cut of one bottom (e.g., 2 × 30 cm = 2 bottoms of 30 cm each).
- **Types of ploughing:** gathering (clockwise, starts at centre), casting (anticlockwise, starts at sides), continuous (one-way/reversible plough — no back furrows or dead furrows, ideal for hillsides and irrigated lands).
- **Specific/unit draft** = draft ÷ furrow cross-sectional area (kgf/cm² or N/cm²).

- Typical unit draft: light sandy soils $\sim 0.2\text{--}0.4 \text{ kgf/cm}^2$, heavy clays $0.7\text{--}1.4 \text{ kgf/cm}^2$ (Kepner: draft varies roughly with speed as $D = a + bV^2$).

The mathematical relationship is modeled as:

$$D = a + bV^2$$

- **(D) (Draft):** The total horizontal force exerted in the direction of travel to pull the implement (measured in (kN) or (lbf)).
- **(V) (Velocity/Speed):** The forward operating speed of the tractor and tool (measured in (km/h) or (mph)).
- **(a) (Static Component):** A constant representing the baseline soil resistance at near-zero speed. It depends heavily on implement type, soil texture, compaction, and depth of cut.
- **(b) (Dynamic Component):** A constant multiplier for the speed-squared term, accounting for the dynamic forces required to accelerate the sheared soil.

2.3 Disc Plough

Cuts, partially inverts and mixes the soil; **no landside in standard disc plough** (side thrust taken by rear furrow wheel).

- **Standard disc plough:** individually mounted concave discs, 60–90 cm diameter, 6–10 mm thick.
 - **Disc angle:** angle between disc face plane and direction of travel = **42–45°** (increasing disc angle → wider cut, better penetration in hard soil).
 - **Tilt angle:** angle between disc plane and vertical = **15–25°** (increases penetration in hard soils).
 - Penetration depends on weight (unlike suction in MB plough) — hence discs carry added ballast; typical weight 180–540 kg per disc bottom equivalent.
- **Vertical disc plough (harrow plough/one-way disc):** discs on a common gang axle, 50–60 cm discs, shallow (8–10 cm) — wheatland plough.
- **Advantages over MB:** works in sticky, hard, stony, rooty soils and in soils where scouring is a problem; rolls over obstacles.
- **Disadvantages:** incomplete inversion, leaves cloddy surface, high weight requirement, rear furrow wheel needed.

2.4 Chisel Plough, Subsoiler, Rotavator

- **Chisel plough:** breaks hard pan/soil to 30–45 cm with narrow curved shanks; no inversion; row spacing of shanks 30 cm typical.
- **Subsoiler:** heavier standard, shatters impervious hardpan at **45–90 cm** depth; requires 60–100 hp; winged subsoilers increase shatter width; **mole plough** forms drainage channels (mole drains) in clay soils.
- **Rotavator (rotary tiller):** PTO-driven blades (L-type most common in India for trashy conditions; also C and J types) on a horizontal shaft; cuts and pulverizes in a single pass — seedbed in one operation; **negative draft (bite length** = forward travel per blade strike) determines pulverization; power consumption high; excellent for puddling.
 - Blade tip speed 4–9 m/s; rotor speed 200–350 rpm typical.
- **Puddlers/puddling:** breaking soil aggregates in standing water to create impervious layer for paddy — cage wheel + rotavator or peg-type puddler.

2.5 Secondary Tillage & Land Development

- **Disc harrow:** gangs of concave discs (35–70 cm); types — single action, **tandem (double action)**, **offset** (works offset to tractor centre-line — orchards), plough harrow. Gang angle 15–30°. Penetration increased by increasing gang angle/adding weight.
- **Cultivators:** rigid tine, **spring-loaded tine** (stony soils — trip and reset), duck-foot (sweep) cultivator for shallow weed-kill and moisture conservation.
- **Blade harrow (bakhar):** indigenous implement — cuts weeds below surface without inversion, retains residue mulch.
- **Levellers/scrapers:** bund former, ridger (ridging body = two opposed mouldboards), **laser-guided land leveller** (± 2 cm accuracy; saves 20–30% irrigation water; components: laser transmitter, receiver, control box, hydraulic valve, drag bucket).
- **Rollers/pulverizers:** clod crushers, plankers for compacting/firming seedbed.

2.6 Draft, Pull and Power

- **Pull** = total force exerted by tractor on implement (along line of hitch).
- **Draft** = horizontal component of pull, in direction of travel: **Draft = Pull $\times$ cos θ** .
- **Unit draft** = draft per unit cross-sectional area of furrow.
- **Drawbar power (kW) = Draft (kN) $\times$ Speed (m/s)**; in field units: DBP (hp) = (Draft, kgf $\times$ Speed, m/s)/75.
- Typical speed of primary tillage 4–6 km/h; secondary 5–8 km/h.